ECE306 - Electronic Circuits And Systems Laboratory
Final Project
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# 4 Analysis and Research Ideas

## 4.1 Concept Overview

The Temperature-Controlled Cooling Fan System was designed to provide an analog solution for monitoring the temperature of an electronic device and responding when overheating conditions occur. Electronic systems like computers and power electronics generate significant heat during operation and excessive temperature can lead to reduced performance or damage of components. The primary goal of this project was to design a circuit capable of sensing temperature changes, by using that information to simulate the function of a cooling fan.

The circuit operates entirely in the analog domain, relying on passive components and MOSFET switches to implement this through voltage threshold behavior. When the sensed temperature exceeds this limit, the circuit will simulate the activation of a cooling fan by illuminating a red LED to indicate this possible overheating. Under normal conditions, a green LED remains illuminated, and indicates that the system temperature is within a safe range. This approach provides intuitive visual feedback regarding system status.

## 4.2 Function Description

The overall circuit consists of three stages: a temperature sensing stage, a control and logic stage and the output stage. These stages work together to translate temperature changes into visual indication. The temperature sensing stage detects changes in ambient temperature (driven by human touch) using a thermistor.

The control stage evaluates this sensed signal and determines whether the temperature exceeds the desired threshold. And the output stage drives the LEDs using MOSFETs.

This design allows each stage to be tested and verified independently. With this structure it is possible to improve reliability and simplify troubleshooting during testing of the project.

## 4.3 Circuit Operation

### 4.3.1 Temperature Sensing

The temperature sensing phase uses a NXRT15WB473FA1B030 thermistor, which is a negative temperature coefficient device. This component has a nominal resistance of 47 kΩ at 25°C and a typical B-constant of approximately 4300 K. These characteristics make the thermistor well suited for detecting temperature variations within the typical operating range of

electronic devices, which generally fall between 30°C and 80°C. This means that it will exhibit a decrease in resistance as temperature increases, this relationship is modeled by:

$$R(t) = (47\ k\Omega) \times e^{B(\frac{1}{t}-\frac{1}{T})}$$, where B = 4300 K and T= 298.15 K

The thermistor is connected in a voltage divider configuration with a fixed resistor and a DC supply voltage. As temperature increases, the resistance of the thermistor decreases, causing the output voltage of the divider to increase. And when the temperature decreases the resistance increases and the output voltage drops.

The output voltage of the divider can be expressed as:

$$V_{out} = V_{supply} \times \frac{R_f}{R_f + R_{therm}}$$

This relationship allows for the selection of resistor values that match the desired input for the MOSFET operating ranges.

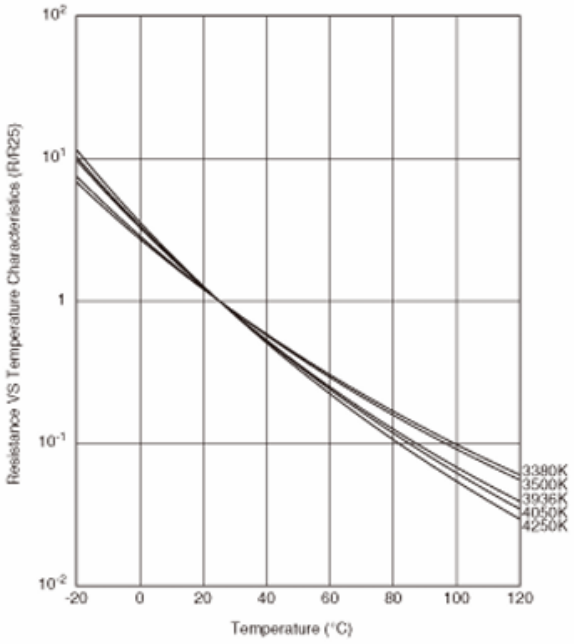



Figure 1. Datasheet Resistance vs. Temperature

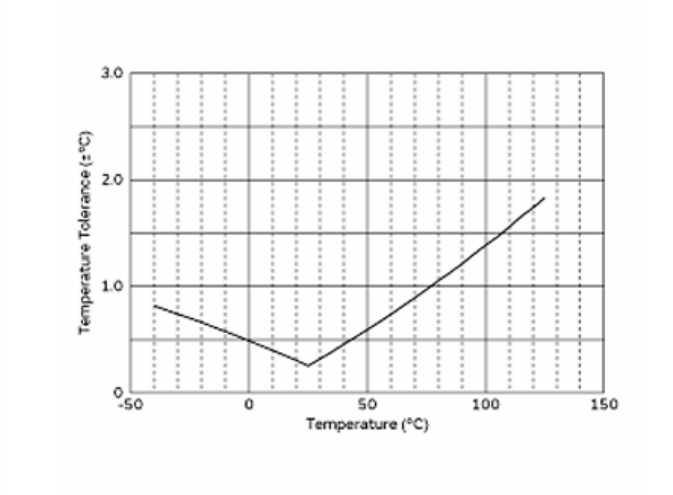



Figure 2. Datasheet Temperature Tolerance.

### 4.3.2 Control and Logic (Red LED)

This stage of the circuit uses an IRFB7545PBF N-channel MOSFET to implement switching using threshold values. The IRFB7545PBF N-channel MOSFET was chosen as the primary switching device due to its low gate threshold voltage range of approximately 2–4 V and its extremely low drain-to-source on-resistance. This configuration relies on the MOSFET's gate to source threshold voltage to determine when the temperature is high enough to activate the cooling system.

As the temperature increases, the voltage produced by the thermistor divider rises and is applied to the gate of the IRFB7545PBF. When this gate voltage exceeds the threshold voltage, then the device transitions from the cutoff region into the triode region. Once fully there, the MOSFET allows current to flow through the load, which includes the green LED stage of the system and the red LED indicator.

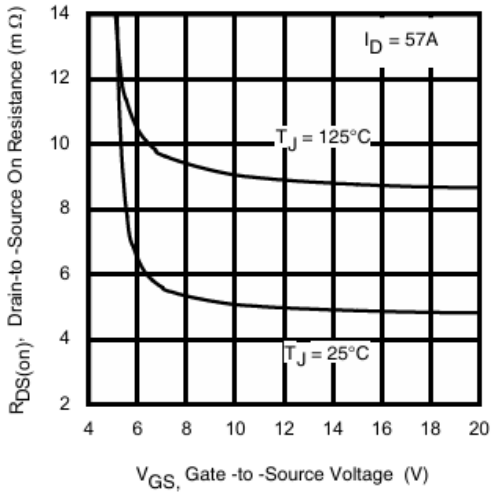



Figure 3. Datasheet Drain Current vs. Gate Voltage.

### 4.3.3 Inversion Logic (Green LED)

To provide clear visual feedback for normal and overheating conditions, a NOT gate was implemented using a p-channel IRF9Z24N MOSFET in combination with a bipolar junction transistor. This allows for a green LED to be normally ON when the system is operating within safe temperature limits and OFF when overheating is detected.

When the temperature is below the threshold, the n-channel MOSFET remains OFF, preventing current flow to this stage of the system. In this stage, the p-channel MOSFET is biased such that the green LED remains turned ON. When temperature rises, then the n-channel turns ON and this activates the green LED. This complementary behavior ensured that there's only one LED active at any given time.

## 4.4 Prototype

Figure 4. Circuit Schematic.

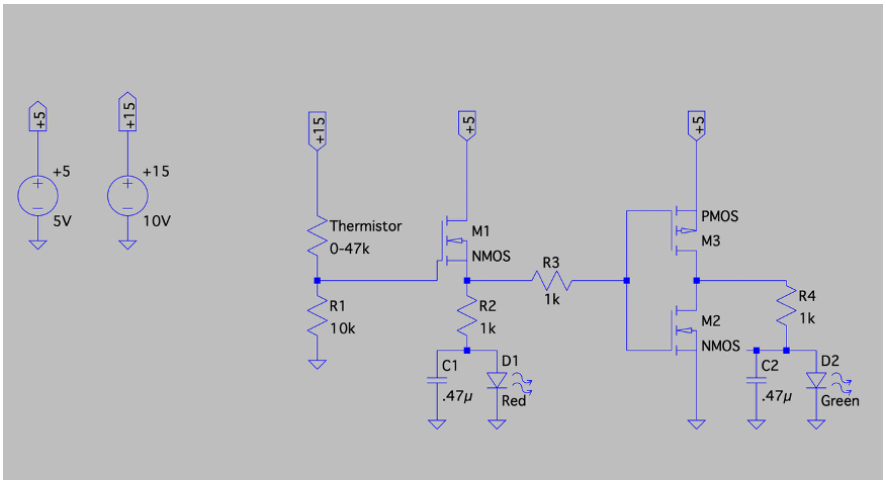



Figure 5. Photograph of Final Prototype.

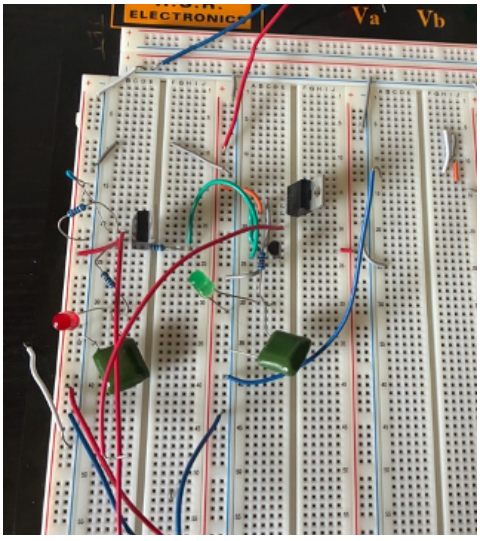


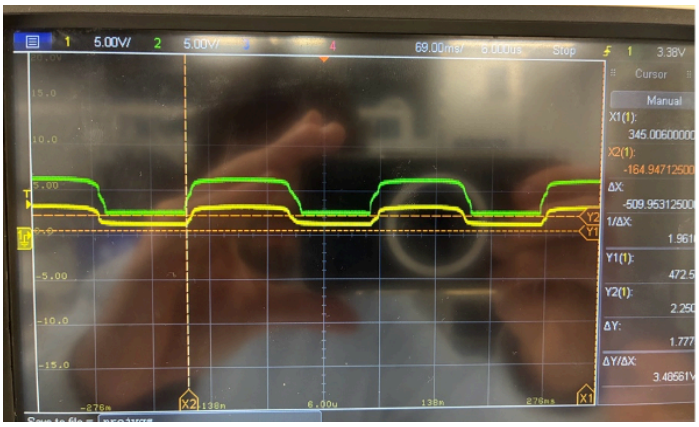
Table 1. List of Components.

1. Thermistor: NXRT15WB473FA1B030
2. M1: NMOS IRFB7545PBF
3. M2: 2N7000
4. M3: 942-IRF9Z24NPBF
5. C1, C2: 0.47uF Capacitor
6. R1, R2: 10kΩ Resistor
7. R3, R4: 1kΩ Resistor

## 4.5 Testing and Verification

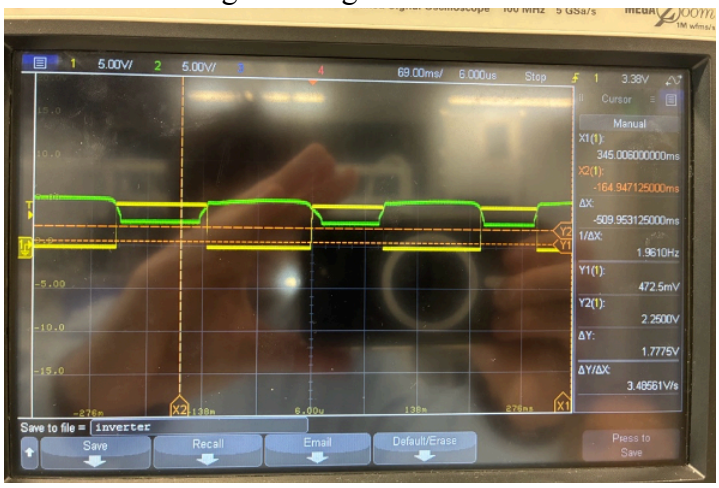
Testing was performed by gradually increasing the temperature applied to the thermistor and monitoring the divider output voltage, MOSFET Vgs, and load response. The measured switching temperature matched those predicted using the datasheets. However there are some slight variations given component tolerances. But the LED indicators behaved as expected.

Figure 6. Experimental vs. Theoretical Results.




In this scope shot, the Vgs going into M1 can be observed. Vgs is at an idle voltage of 0.75V, (2.25 – 1.5). Once the thermistor is touched Vg climbs to ~6V, and Vs climbs to ~2V, which leads to a Vgs of 4V, well over the necessary 3.7 to be in saturation.

Figure 7. Vg into Inverter




In this scope shot, the Vg going into M1 can be observed as well as the Vd on the inverter circuit. The drain voltage is what will cause current flow through the green LED and it is clearly visible that as the Vg decreases, or the thermistor gets colder, the signal into the LED will turn on and drive the green status LED.

## 4.6 Conclusion

In conclusion, the designed sensing circuit successfully integrates temperature sensing, analog signal conditioning, and MOSFET-based switching to provide automatic thermal protection. The use of datasheet parameters and analytical calculations ensured predictable behavior and efficient operation. Experimental results closely matched theoretical expectations, validating the design approach and demonstrating a practical application of sensing circuits covered in the ECE 306 lab.

## 5 Citations


*NXRT15WB473FA1B030 | DigiKey Electronics*. (2025). DigiKey Electronics. https://www.digikey.com/en/products/detail/murata-electronics/NXRT15WB473FA1B030/4422435?gclid=ae681492100d13f70b687dac0a36d1b1&gclsrc=3p.ds&msclkid=ae681492100d13f70b687dac0a36d1b1

*IRFB7545PBF | DigiKey Electronics*. (2025). DigiKey Electronics. https://www.digikey.com/en/products/detail/infineon-technologies/IRFB7545PBF/4772483?gclid=8136c1e1d47d1797011ba8dfce957b8d&gclsrc=3p.ds&msclkid=8136c1e1d47d1797011ba8dfce957b8d